Overview of FIR and IIR Filters

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Key Concepts

1) A finite impulse response (FIR) filter has an impulse response of limited duration.

    a) The difference equation for an FIR filter does not involve recursion: past outputs have no influence on present outputs.

    b) FIR filters are sometimes called "all zero" because the poles are always located at $z = 0$.

2) The impulse response of an infinite impulse response (IIR) filter lasts forever.

a) The difference equation for an IIR filter has recursion: past outputs affect present outputs. This is why the impulse response never terminates.

b) Both pole and zero locations can be chosen with IIR filters.

3) FIR filter design characteristics -

a) Optimization methods can be used to find the "best" filter for a set of specifications.

b) The magnitude and phase response can be specified independently.

c) Can require FIR filters to have linear phase.

d) FIR designs can require large order.

4) IIR filter design characteristics -

a) Adapt continuous-time filter designs.

b) Usually only magnitude is specified. Phase is usually not controlled and is always nonlinear.

c) Often uses fewer coefficients than FIR for comparable magnitude response specifications.

Video Screenshots

Two categories of discrete-time filters

Finite Impulse Response FIR

$$y[n] = \sum_{k=0}^{M} b_k x[n-k]$$

$$H(z) = \sum_{k=0}^{M} b_k z^{-k}$$

$$= b_0 \prod_{k=1}^{M} (1 - c_k z^{-1})$$

$$h[n] = \begin{cases} b_n, & 0 \le n \le M \\ 0, & \text{otherwise} \end{cases}$$

Infinite Impulse Response IIR

$$y[n] = \sum_{k=0}^{M} b_k x[n-k] - \sum_{k=1}^{N} a_k y[n-k]$$

$$H(z) = \frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}}, \quad a_0 = 1$$

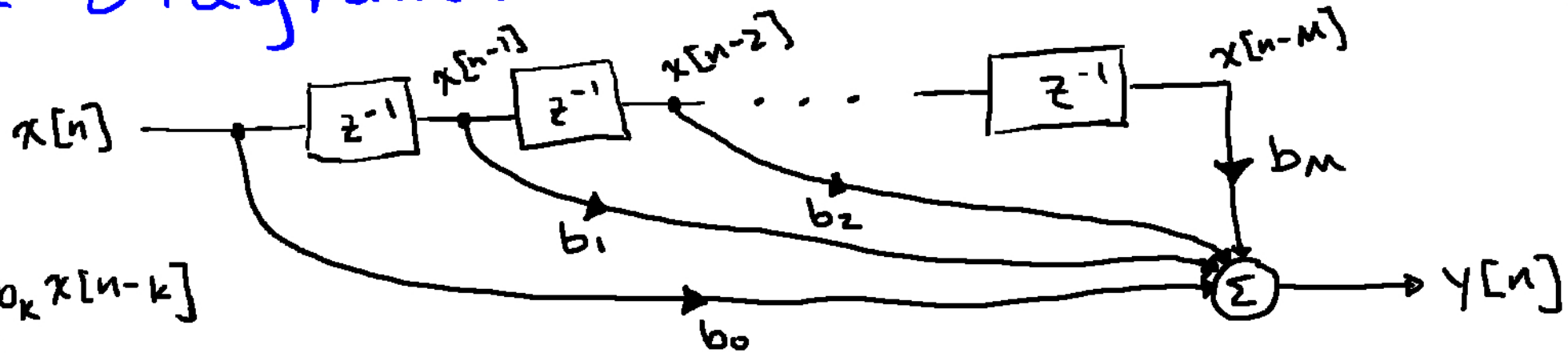
$$= \frac{b_0 \prod_{k=1}^{M} (1 - c_k z^{-1})}{\prod_{k=1}^{N} (1 - d_k z^{-1})}$$

$$h[n] = \sum_{k=1}^{N} A_k (d_k)^n u[n]$$

Block Diagrams

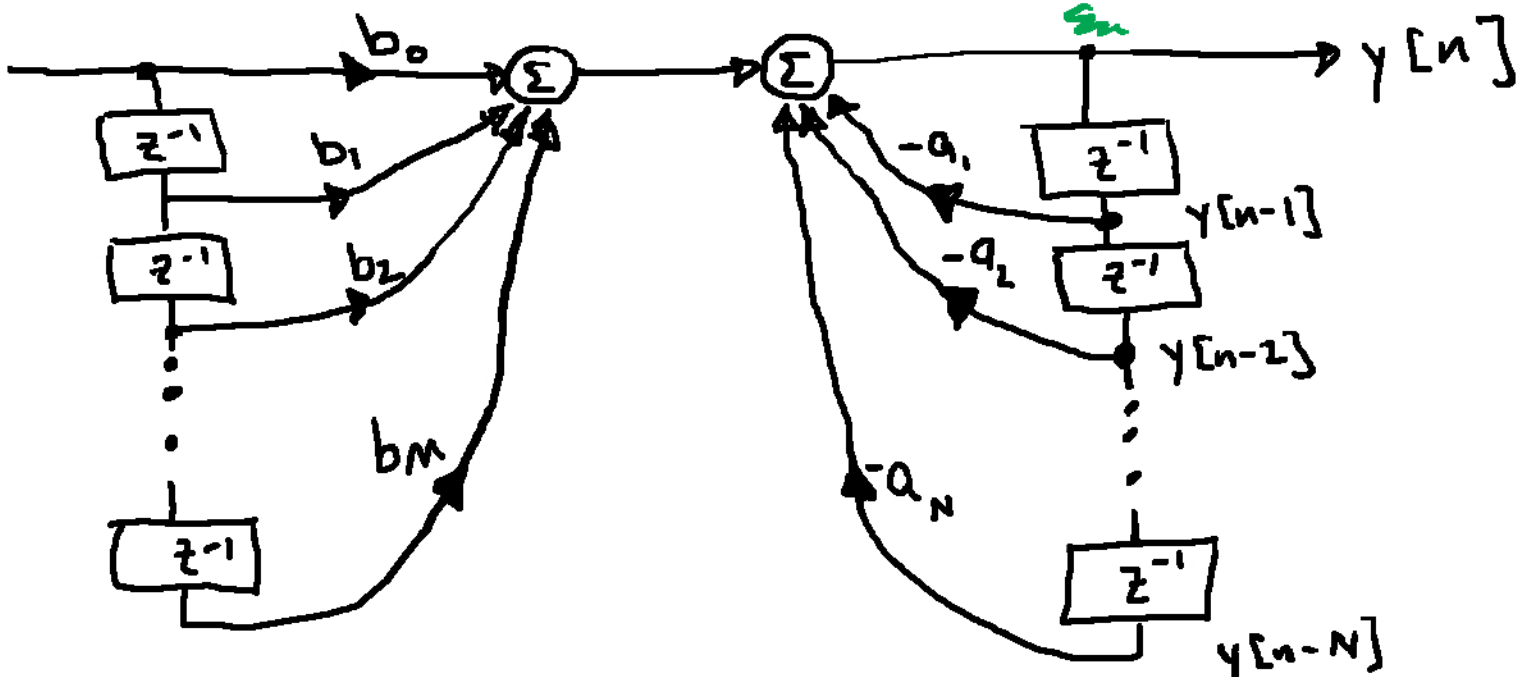
FIR

$$y[n] = \sum_{k=0}^{M} b_k x[n-k]$$



IIR

$$y[n] = \sum_{k=0}^{M} b_k x[n-k] - \sum_{k=1}^{N} a_k y[n-k]$$



Filter Designs

Desired frequency response - $H_d(e^{j\omega})$

FIR

$$H(e^{j\omega}) = \sum_{k=0}^{M} b_k e^{-jk\omega}$$

- Optimization based design
- Arbitrary magnitude/phase response
- Can obtain linear phase
- Can require large M

IIR

$$H(e^{j\omega}) = \frac{\sum_{k=0}^{M} b_k e^{-jk\omega}}{\sum_{k=0}^{N} a_k e^{-jk\omega}}, \quad a_0 = 1$$

- Transform analog filter designs
- Frequency selective gain
- Nonlinear phase, no control
- Fewer coefficients than FIR

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